

Likelihood-Ranked Selection in Self-Consuming Diffusion: A W_2 Negative Result and a Narrow Minority-Retention Regime

Muhammad Alamzeb
Independent researcher
shayankhanmahar@gmail.com

Abstract

Self-consuming training loops can induce *model collapse*, including loss of rare modes. We study unlabeled likelihood ranking of synthetics in self-consuming DDPM training on imbalanced mixtures. **Primary claim:** selecting by top likelihood does *not* improve sliced W_2 vs. random- k at generation 1 ($\Delta W_2 > 0$ on 23/23 seed-imbalance cells; bootstrap 95% CI of the mean excludes zero; Wilcoxon $p=2.9\times 10^{-5}$, Holm $p=1.4\times 10^{-4}$). **Secondary, narrow-regime observation:** the full bottom>rand>top minority-retention order appears only at moderate imbalance $\rho=5$ ($n=10$ seeds, 10/10; exact Wilcoxon $p=0.002$ for both paired contrasts vs. random- k ; bootstrap CIs exclude zero), and already fails for most seeds at $\rho=7$ (2/5) and $\rho=10$ (3/8). This is a workshop-scale empirical note on controllable mixtures and tiny Digits checks—*not* a result on ImageNet-scale diffusion. Code: <https://github.com/muhammadalamzeb/likelihood-rank-self-consuming-diffusion>.

1 Introduction

As synthetic data proliferates, generative models are increasingly trained on mixtures of real and model-generated samples. Iterated self-consumption can cause *model collapse* [Shumailov et al., 2023, Alemohammad et al., 2023, Gerstgrasser et al., 2024]. Mitigations include accumulating real data [Gerstgrasser et al., 2024] and filtering synthetics [Feng et al., 2024, Cai et al., 2025]. When the only score is the generator’s own likelihood, how should one select synthetics? We isolate an *interventional* selection policy at matched synthetic count and matched real-mix α , with mode-exact metrics.

What is new. A powered falsification of “top- k looks better on aggregate W_2 ,” plus a *narrow-regime* minority-retention observation at moderate imbalance that does *not* generalize to higher ρ or to image-scale models.

Contributions. (1) Matched-budget DDPM protocol with exact mode masses. (2) Primary: top- k does not improve sliced W_2 vs. random- k at generation 1 ($n=23$ cells). (3) Secondary: bottom>rand>top minority retention at $\rho=5$ ($n=10$), unreliable at $\rho=7, 10$. (4) Tiny Digits checks (PCA supports; pixel DDPM / CVAE do not stably replicate the full order).

2 Related work

Shumailov et al. [2023], Alemohammad et al. [2023] document collapse and quality-biased sampling (*MAD*). Gerstgrasser et al. [2024] show accumulating real+synthetic data can avoid collapse. Feng

Table 1: Primary cells ($\text{seed} \times \rho$). Seeds 8–9 only at $\rho=5$.

$\rho \backslash \text{seed}$	0	1	2	3	4	5	6	7	8	9
5	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
7	✓	✓	✓	✓	✓	—	—	—	—	—
10	✓	✓	✓	✓	✓	✓	✓	✓	—	—

et al. [2024] study task verification of labeled synth; Cai et al. [2025] (*LSF*) filter via latent probes. Qiao et al. [2026] prove that selection against a *biased local* real-data reference can prune global tail modes and accelerate collapse in siloed settings; their remedy uses collaborative Wasserstein proxy references. We study a complementary failure mode: ranking synthetics by the *generator’s own* denoising-likelihood proxy, with a matched-budget top- k / random- k / bottom- k contrast and minority-mass retention as the primary diversity metric. Inference-time temperature methods [Xu et al., 2025, Li et al., 2026] reshape sampling diversity; we intervene on training-set selection.

3 Method

3.1 Self-consuming loop

Train DDPM $f^{(g)}$ on $\mathcal{D}^{(g)}$, sample synthetics, mix fixed real buffer (fraction α) with selected synthetics.

3.2 Likelihood proxy and policies

Score $s(x)$: mean denoising MSE over a t -grid (lower = higher likelihood).

- **top- k / bottom- k / random- k :** pool of size $k = \max(n_{\text{syn}}, \lfloor k_{\text{frac}} N \rfloor)$ by lowest / highest / uniform scores; then draw n_{syn} and mix with α real.
- **mix:** same α , but synthetic portion is a uniform subsample of all N synthetics (no rank filter).
- **replace:** pure synthetic.

Thus mix $\neq$ random- k .

3.3 Metrics

Nearest-mean mode masses; minority-mean m ; ratio $r = m^{(1)}/m^{(0)}$ (“retention,” allowed > 1). Sliced W_2 : mean squared 1D Wasserstein over **24** projections; ≤ 400 synth and ≤ 400 held-out real points. Hypothesis tests: bootstrap 95% CIs of the mean paired Δ ; Wilcoxon signed-rank with **exact** sign enumeration for $n \leq 16$ (midranks for ties) and a normal approximation for larger n .

4 Experiments

4.1 Primary design ($n=23$ cells)

$K=4$, $G=5$, $\alpha=0.5$, $k_{\text{frac}}=0.5$. Logs: `w2_fs.jsonl`. Matrix: $\rho=5$ with $n=10$ seeds (escapes the $n=5$ exact- p floor of 0.0625), intermediate $\rho=7$ ($n=5$), and $\rho=10$ ($n=8$).

Table 2: $\Delta W_2(\text{top} - \text{rand})$ by generation on all 23 cells (positive = top worse). Bootstrap 95% CI; Wilcoxon p ; Holm across five generation-wise tests.

g	Mean	Std	d_z	95% CI	$\#\{> 0\}$	p (Holm)
1	+0.089	0.077	1.16	[0.060, 0.121]	23/23	2.9×10^{-5} (1.4×10^{-4})
2	+0.046	0.044	1.04	[0.029, 0.065]	21/23	0.00015 (0.00061)
3	+0.034	0.030	1.14	[0.022, 0.045]	21/23	0.00022 (0.00066)
4	+0.021	0.033	0.62	[0.007, 0.034]	19/23	0.0013 (0.0027)
5	+0.020	0.027	0.73	[0.010, 0.031]	19/23	0.0015 (0.0027)

Table 3: Per- ρ mean r and seed-wise counts (generation 1). Full order = bottom>rand>top.

ρ	n	bottom	rand	top	Full order	top<rand / bot>rand
5	10	0.651	0.366	0.141	10/10	10/10 / 10/10
7	5	0.437	0.266	0.145	2/5	4/5 / 3/5
10	8	0.329	0.247	0.164	3/8	7/8 / 4/8

4.2 Sliced W_2 (primary powered claim)

Generation-wise tests are repeated measures (serially correlated); Holm is conservative. Evidence against top- k improving W_2 is strongest at $g=1$ (CI excludes zero; unanimous sign).

4.3 Retention: stratified secondary analysis

Because strata differ, we do **not** treat a pooled Wilcoxon as the primary claim. We report bootstrap 95% CIs alongside Wilcoxon p . Table 3 gives means and seed-wise order counts; Table 4 gives stratified paired contrasts.

At $\rho=5$ ($n=10$), all seeds agree and exact $p=0.002$ clears $\alpha=0.05$ (Holm-adjusted within the two contrasts: $p=0.0039$). We still frame this as a **narrow-regime** finding: the full order holds on only 2/5 seeds at $\rho=7$ and 3/8 at $\rho=10$. Top-rand CIs exclude zero at $\rho=7$ and $\rho=10$, but bottom-rand’s CI includes zero at $\rho=10$. Pooled $n=23$ tests are secondary (mix heterogeneous strata).

4.4 Real-mix α ablation (illustrative; $n=2$)

At $\alpha=0.5$, seed 0 breaks the full order (bottom $m^{(1)}$ slightly below random) while seed 1 holds—hence 1/2 (means still rank bottom>rand>top). At $\alpha=0.25$, likewise 1/2; at $\alpha=0.75$, top vs. random reverses on average (0/2).

4.5 Transfer checks (directional; $n=3$)

Minimum two-sided exact Wilcoxon p at $n=3$ with all signs agreeing is 0.25; we report Cohen’s d_z against that bound and do not claim powered generalization.

- **8D GMM:** 0.55 ± 0.20 / 0.27 ± 0.06 / 0.08 ± 0.03 ; order 3/3; top-rand $d_z = -2.56$, bottom-rand $d_z = +1.28$ (exact $p=0.25$).
- **Digits-PCA:** 1.49 ± 0.11 / 1.11 ± 0.08 / 0.68 ± 0.17 ; order 3/3; top-rand $d_z = -1.72$, bottom-rand $d_z = +2.67$ (exact $p=0.25$).

Table 4: Stratified paired retention contrasts vs. random- k (bootstrap 95% CI of mean Δ ; exact Wilcoxon for $n \leq 16$).

ρ	Contrast	Mean Δ	d_z	95% CI	Exact p
5	top-rand	-0.225	-1.82	$[-0.30, -0.15]$	0.0020
5	bottom-rand	+0.285	+2.08	$[0.21, 0.37]$	0.0020
7	top-rand	-0.121	-1.09	$[-0.21, -0.04]$	0.125
7	bottom-rand	+0.171	+0.92	$[0.03, 0.32]$	0.250
10	top-rand	-0.083	-0.79	$[-0.14, -0.01]$	0.078
10	bottom-rand	+0.082	+0.44	$[-0.03, 0.21]$	0.578



Figure 1: Retention r stratified by ρ (narrow regime at $\rho=5$).

- **Digits pixel-space DDPM (8×8 MLP denoiser):** order 1/3 at $\rho=5$; means 1.25 / 0.88 / 1.23; bottom>rand on 3/3 seeds ($d_z=+1.27$) but top vs. random is inconsistent (exact signs 1/3 for top<rand). A small image-diffusion check, not ImageNet-scale validation; full ordering not replicated.
- **Digits-CVAE (inconclusive):** pixel-space conditional VAE did not reproduce a stable bottom>rand>top order.

4.6 Selection-pool size k_{frac} (illustrative; $n=2$)

Top- k remains worst on means; at 0.75, bottom vs. random flips (0/2 full order).

Table 5: Mean r by α (seeds $\{0, 1\}$, $\rho=5$, default $k_{\text{frac}}=0.5$). “Order holds” = bottom>rand>top on $g=1$ minority mass. The $\alpha=0.5$ row is the *same cell* as $k_{\text{frac}}=0.5$ in Table 6.

α	bottom	rand	top	Order holds
0.25	0.25	0.17	0.05	1/2
0.50	0.62	0.49	0.24	1/2
0.75	1.34	0.77	0.94	0/2

Table 6: k_{frac} ablation (seeds $\{0, 1\}$, $\rho=5$, $\alpha=0.5$). $0.25 \equiv 0.5$ when n_{syn} binds. The $k_{\text{frac}}=0.5$ row duplicates Table 5’s $\alpha=0.5$ row.

k_{frac}	bottom	rand	top	Order holds
0.25	0.62	0.49	0.24	1/2
0.50	0.62	0.49	0.24	1/2
0.75	0.48	0.61	0.22	0/2

5 Analysis

Top- k retains high-likelihood (typically majority) synthetics. Aggregate sliced W_2 is a poor monitor of minority survival: top- k is consistently *worse* on W_2 at $g=1$ across the full matrix, while minority retention shows a *narrow* bottom>rand>top regime only at $\rho=5$. That regime already fails for most seeds at $\rho=7$ and $\rho=10$.

6 Limitations

Scale. All primary results are on 2D/8D GMMs and tiny Digits (8×8) checks. We have *not* demonstrated these effects on any real image-scale diffusion model (e.g. CIFAR/ImageNet UNet or DiT); Digits pixel-space DDPM does not stably replicate the full retention order (1/3). **Scope.** Workshop / short empirical note—not a definitive scaling result. α and k_{frac} ablations use two seeds; their default rows are the same experimental cell. **Multiplicity:** Holm within the W_2 generation family and within the two $\rho=5$ retention contrasts; other tests descriptive. Complementary to MAD/LSF/Feng and siloed-selection theory [Qiao et al., 2026].

Exploratory search. The retention hypothesis is the survivor of an earlier exploratory search (historical notes under `docs/` in the repository, not part of the submission narrative). We mitigate overclaiming by making W_2 the primary claim, stratifying retention by ρ , demoting pooled tests, and leading with bootstrap CIs.

Note on double-blind review. The public GitHub URL reveals author identity. For double-blind venues, provide an anonymous code mirror; for arXiv, the named repository is appropriate.

7 Conclusion

Top- k likelihood selection does not improve sliced W_2 vs. random- k at generation 1 in our matrix. A bottom>rand>top minority-retention order appears in a **narrow** moderate-imbalance regime

($\rho=5$) and should not be stated as a general law. Preferring “most likely” synthetics is not a free lunch on either aggregate W_2 or rare-mode survival when minorities remain measurable.

References

- Sina Alemohammad, Josue Casco-Rodriguez, Lorenzo Luzi, Ahmed Imtiaz Humayun, Hossein Babaei, Daniel LeJeune, Ali Siahkoohi, and Richard G. Baraniuk. Self-consuming generative models go MAD, 2023.
- Zhongteng Cai, Yaxuan Wang, Yang Liu, and Xueru Zhang. Stabilizing self-consuming diffusion models with latent space filtering, 2025.
- Yunzhen Feng, Elvis Dohmatob, Pu Yang, Francois Charton, and Julia Kempe. Beyond model collapse: Scaling up with synthesized data requires verification, 2024.
- Matthias Gerstgrasser, Rylan Schaeffer, Apratim Dey, Rafael Rafailov, Dhruv Pai, Henry Sleight, John Hughes, Tomasz Korbak, Rajashree Agrawal, Andrey Gromov, Daniel A. Roberts, Diyi Yang, David Donoho, and Sanmi Koyejo. Is model collapse inevitable? breaking the curse of recursion by accumulating real and synthetic data, 2024.
- Peizhuo Li, Emre Aksan, Alexandru-Eugen Ichim, Thabo Beeler, and Olga Sorkine-Hornung. Diversify diffusion with temperature sampling and variance-corrective time shifting, 2026.
- Xinbao Qiao, Xianglong Du, Wei Liu, Jingqi Zhang, Peihua Mai, Meng Zhang, and Yan Pang. When sample selection bias precipitates model collapse, 2026. ICML 2026.
- Ilya Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. The curse of recursion: Training on generated data makes models forget, 2023.
- Yanbo Xu, Yu Wu, Sungjae Park, Zhizhou Zhou, and Shubham Tulsiani. Temporal score rescaling for temperature sampling in diffusion and flow models, 2025.

A Reproducibility

`python scripts/reproduce_w2.py`; stats tables via `experiments/scripts/make_submission_stats.py`.
Code: <https://github.com/muhammadalamzeb/likelihood-rank-self-consuming-diffusion>.